

ASSIGNMENT

Question. What is covariance, and what does it measure?

Question. What are the possible ranges of covariance values?

Question. How is correlation different from covariance?

Question. What does a correlation coefficient of 0 indicate?

Question. If the correlation coefficient is -0.75, what does it suggest about the relationship between two variables?

Question. Define probability.

Question. What is marginal probability?

Question. How is conditional probability calculated?

Question. What does $P(A|B)$ represent in conditional probability?

Question. What is the Multiplication Rule of Probability?

Question. Describe the characteristics of a normal distribution.

Question. What is the Central Limit Theorem?

Question. What is hypothesis testing?

Question. Define the null hypothesis (H_0).

Question. Define data skewness.

Question. What is the mode in a dataset?

Question. Calculate the median of the following dataset: [12, 15, 18, 22, 27].

Question. What is the range of a dataset?

Question. How is the variance of a dataset calculated?

Question. Define descriptive statistics.

Question. Name the measures of central tendency.

Question. Calculate the mean of the following dataset: [10, 15, 20, 25, 30].

Question. What is the interquartile range (IQR) of a dataset?

Question. How is the standard deviation different from variance?

Question. Calculate the covariance between two variables X and Y for the following data:

X: [10, 15, 20, 25, 30]

Y: [5, 12, 18, 22, 28]

Question. Calculate the correlation coefficient (Pearson's r) for the data in prev question.

Question. Given a covariance of 45 and standard deviations of $X = 7$ and $Y = 9$, calculate the correlation coefficient (r) between X and Y .

Question. If the correlation coefficient between two variables X and Y is -0.6 and the standard deviation of X is 8, what is the standard deviation of Y ?

Question. Two variables, A and B , have a correlation coefficient of 0.85 . If the standard deviation of A is 10, what is the standard deviation of B ?

Question. A bag contains 5 red balls, 4 blue balls, and 3 green balls. If you draw one ball at random, what is the probability that it is not green?

Question. Given $P(A) = 0.4$, $P(B) = 0.6$, and $P(A \cap B) = 0.2$, calculate $P(A|B)$ using conditional probability.

Question. In a deck of 52 cards, what is the probability of drawing a heart or a spade?

Question. A box contains 10 red balls, 8 blue balls, and 6 green balls. If you draw two balls without replacement, what is the probability of getting one red and one blue ball in either order?

Question. If two dice are rolled, what is the probability of getting a sum of 7?

Question. You are conducting a hypothesis test with a significance level (α) of 0.05 . Your sample gives you a test statistic (z) of 2.5 . Should you reject the null hypothesis?

Question. In a factory, the weights of boxes of chocolates follow a normal distribution with a mean of 200 grams and a standard deviation of 20 grams. What is the probability that a randomly selected box weighs less than 180 grams?

Question. The lifetimes of light bulbs follow a normal distribution with a mean of 1000 hours and a standard deviation of 100 hours. What percentage of bulbs can be expected to last between 900 and 1100 hours?

Question. The heights of students in a school are normally distributed with a mean of 160 cm and a standard deviation of 10 cm. What is the height below which 90% of the students fall?

Question. Calculate the skewness of the following dataset: [5, 10, 15, 20, 15, 30, 35, 40].

Question. Find the mode of the dataset: [5, 10, 15, 20, 15, 30, 35, 40].

Question. Calculate the median of the dataset: [7, 14, 21, 28, 35, 42].

Question. Calculate the range of the following dataset: [12, 20, 30, 8, 17].

Question. Calculate the variance of the dataset: [3, 6, 9, 12, 15].

Question. Calculate the standard deviation of the dataset in question 20.

Question. Find the first quartile (Q1) and third quartile (Q3) for the dataset: [5, 8, 10, 14, 20, 27].

Question. Calculate the interquartile range (IQR) for the dataset in question 22.

Question. Calculate the coefficient of variation (CV) for a dataset with a mean of 40 and a standard deviation of 8.

Question. Calculate the covariance between two variables X and Y for the following data:

X: [8, 12, 15, 20, 25]

Y: [10, 18, 20, 24, 30]

Question. Calculate the correlation coefficient (Pearson's r) for the data in question 26.

Question. Given a covariance of -30 and standard deviations of $X = 6$ and $Y = 9$, calculate the correlation coefficient (r) between X and Y.

Question. If the correlation coefficient between two variables X and Y is 0.2 and the standard deviation of X is 12, what is the standard deviation of Y ?

Question. Two variables, A and B , have a correlation coefficient of -0.65. If the standard deviation of A is 15, what is the standard deviation of B ?

Question. A bag contains 5 red balls, 4 blue balls, and 3 green balls. If you draw one ball at random, what is the probability that it is blue or green?

Question. Given $P(A) = 0.3$, $P(B) = 0.6$, and $P(A \cap B) = 0.1$, calculate $P(A|B)$ using conditional probability.

Question. In a deck of 52 cards, what is the probability of drawing a face card (jack, queen, king) or a red card (hearts or diamonds)?

Question. A box contains 12 marbles: 4 red, 3 green, and 5 blue. If you draw three marbles without replacement, what is the probability of getting at least one red marble?

Question. If three dice are rolled, what is the probability of getting a sum of 12?

Question. The scores on a standardized test follow a normal distribution with a mean (μ) of 600 and a standard deviation (σ) of 75. What is the probability that a randomly selected student scored between 525 and 675?

Question. You are conducting a hypothesis test with a significance level (α) of 0.01. Your sample gives you a test statistic (z) of -2.75. Should you reject the null hypothesis?

Question. In a factory, the weights of boxes of chocolates follow a normal distribution with a mean of 250 grams and a standard deviation of 20 grams. What is the probability that a randomly selected box weighs more than 270 grams?

Question. The lifetimes of batteries follow a normal distribution with a mean of 400 hours and a standard deviation of 50 hours. What is the 90th percentile of battery lifetimes?

Question. The heights of adult women in a city are normally distributed with a mean of 165 cm and a standard deviation of 8 cm. What is the probability that a randomly selected woman is taller than 175 cm?

Question. Calculate the skewness of the following dataset: [8, 12, 15, 20, 25, 35, 40, 45, 50].

Question. Find the mode of the dataset: [10, 15, 20, 20, 25, 30, 30, 35, 40].

Question. Calculate the median of the dataset: [7, 14, 21, 28, 35, 42, 49, 56, 63].

Question. Calculate the range of the following dataset: [12, 20, 30, 8, 17, 45, 62, 38].

Question. Calculate the variance of the dataset: [5, 10, 15, 20, 25, 30, 35, 40, 45].

Question. Calculate the standard deviation of the dataset in question 45.

Question. Find the first quartile (Q1) and third quartile (Q3) for the dataset: [5, 8, 10, 14, 20, 27, 33, 41, 50].

Question. Calculate the interquartile range (IQR) for the dataset in question 47.

Question. Calculate the coefficient of variation (CV) for a dataset with a mean of 60 and a standard deviation of 15.